

ECO 362: Financial Investments

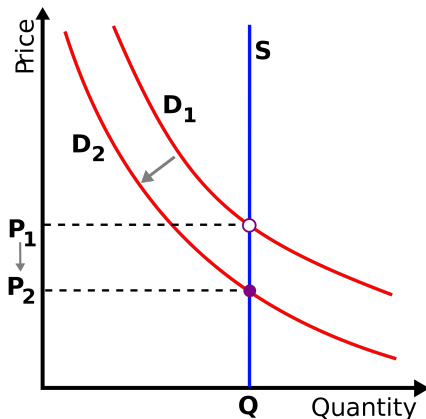
Capital Asset Pricing Model

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Apple Inc. on 8/31/2022

- 16.095 billion shares outstanding.
- Closed at \$157.22 per share.
- Market equity of \$2.554 trillion ($= \157.22×16.095 billion).
- 6.07% of total US stock market.



Relation between price and expected return

- Investors pay P_0 today.
- In the future, they receive a dividend D_1 , and the share price moves to P_1 . The return is

$$1 + R = \frac{P_1 + D_1}{P_0}$$

- The expected return is

$$\mathbb{E}[1 + R] = \frac{\mathbb{E}[P_1 + D_1]}{P_0}$$

- High expected return $\Leftrightarrow$ Low price
- Price of Apple stock adjusts so that investors are collectively happy to hold exactly 16.095 billion shares.

Assets

- Riskless interest rate R_b .
- I risky assets indexed by $i = 1, \dots, I$.
- $\mathbf{R}$ is a vector of returns, whose i th element is R_i .
- Expected returns are $\mathbb{E}[\mathbf{R}]$.
- The covariance matrix is

$$\Sigma = \mathbb{E}[(\mathbf{R} - \mathbb{E}[\mathbf{R}])(\mathbf{R} - \mathbb{E}[\mathbf{R}])']$$

Supply

- Let $\mathbf{w}_m$ be a vector of market weights.
 - The i th element is asset i 's market equity as a share of total market equity.
- The return on the market portfolio is

$$R_m = R_b + \mathbf{w}_m'(\mathbf{R} - R_b\mathbf{1})$$

- Note that $R_m - \mathbb{E}[R_m] = \mathbf{w}_m'(\mathbf{R} - \mathbb{E}[\mathbf{R}])$.

Demand

- N mean-variance investors indexed by $n = 1, \dots, N$.
- Each investor has wealth A_n and risk aversion γ_n .
- Total wealth in the economy is $A = \sum_{n=1}^N A_n$.
 - Also equal to total market equity.
- From the lecture on “Diversification”, we know that

$$\mathbf{w}_n = \frac{1}{\gamma_n} \Sigma^{-1} \mathbb{E}[\mathbf{R} - R_b \mathbf{1}]$$

Market clearing

- Market clearing (supply = demand).

$$\begin{aligned} A\mathbf{w}_m &= \sum_{n=1}^N A_n \mathbf{w}_n \\ &= \sum_{n=1}^N \frac{A_n}{\gamma_n} \Sigma^{-1} \mathbb{E}[\mathbf{R} - R_b \mathbf{1}] \end{aligned}$$

- Dividing by total wealth,

$$\mathbf{w}_m = \frac{1}{\bar{\gamma}} \Sigma^{-1} \mathbb{E}[\mathbf{R} - R_b \mathbf{1}]$$

where $\frac{1}{\bar{\gamma}} = \sum_{n=1}^N \frac{A_n}{A\gamma_n}$.

- Rearrange as

$$\begin{aligned}\mathbb{E}[\mathbf{R} - R_b \mathbf{1}] &= \bar{\gamma} \Sigma \mathbf{w}_m \\ &= \bar{\gamma} \mathbb{E}[(\mathbf{R} - \mathbb{E}[\mathbf{R}])(\mathbf{R} - \mathbb{E}[\mathbf{R}])'] \mathbf{w}_m \\ &= \bar{\gamma} \mathbb{E}[(\mathbf{R} - \mathbb{E}[\mathbf{R}])(R_m - \mathbb{E}[R_m])]\end{aligned}$$

- The i th row of this equation is

$$\mathbb{E}[R_i - R_b] = \bar{\gamma} \text{Cov}(R_i, R_m) \quad (1)$$

- Multiply both sides by $\mathbf{w}_m'$.

$$\begin{aligned}\mathbf{w}_m' \mathbb{E}[\mathbf{R} - R_b \mathbf{1}] &= \bar{\gamma} \mathbf{w}_m' \mathbb{E}[(\mathbf{R} - \mathbb{E}[\mathbf{R}])(R_m - \mathbb{E}[R_m])] \\ \mathbb{E}[R_m - R_b] &= \bar{\gamma} \underbrace{\mathbb{E}[(R_m - \mathbb{E}[R_m])^2]}_{\text{Var}(R_m)}\end{aligned} \quad (2)$$

Capital Asset Pricing Model (CAPM)

- Substitute (2) into (1) to eliminate $\bar{\gamma}$.

$$\mathbb{E}[R_i - R_b] = \beta_i \mathbb{E}[R_m - R_b]$$

where

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

is asset i 's market beta.

- Special cases:
 - For the market portfolio ($R_i = R_m$), $\beta_i = 1$.
 - For idiosyncratic risk ($\beta_i = 0$), $\mathbb{E}[R_i] = R_b$.

Key implications

- Assets with higher market beta have higher expected returns.
- Only systematic risk that is correlated with the market is rewarded.
- Idiosyncratic risk is not rewarded and earns expected return equal to the riskless rate.
- Intuition:
 - Portfolio separation theorem implies that all investors hold a combination of the riskless asset and a diversified portfolio (i.e., the tangency portfolio).
 - No investor bears idiosyncratic risk, so it cannot be rewarded.
 - Investors that bear more market risk are rewarded with higher expected returns.

Using the CAPM for discounting

- An asset costs P_i today and pays a random cash flow X_i at a future date (e.g., in 10 years).
- Its rate of return is

$$1 + R_i = \frac{X_i}{P_i}$$

- Applying the CAPM to this asset,

$$\mathbb{E}[R_i - R_b] = \beta_i \mathbb{E}[R_m - R_b]$$

$$\mathbb{E}\left[\frac{X_i}{P_i}\right] = 1 + R_b + \beta_i \mathbb{E}[R_m - R_b]$$

- Rearrange for a present-value formula.

$$P_i = \frac{\mathbb{E}[X_i]}{1 + R_b + \beta_i \mathbb{E}[R_m - R_b]}$$

Present-value formula

- $R_b + \beta_i \mathbb{E}[R_m - R_b]$ is a risk-adjusted interest rate, which is increasing in β_i .
- For a riskless cash flow with $\beta_i = 0$, the formula simplifies to

$$P_i = \frac{\mathbb{E}[X_i]}{1 + R_b}$$

where R_b is the riskless rate.

- Be careful to match the horizon of returns with the maturity of cash flow.
- For a 10-year asset,
 - $1 + R_b = (1 + Y_b(10))^{10}$ where $Y_b(10)$ is the 10-year zero-coupon yield.
 - $\mathbb{E}[R_m - R_b]$ is the expected excess market return over 10 years.

Testing the CAPM

- Using time-series data on returns, estimate a linear regression for each asset i .

$$R_{i,t} - R_{b,t} = \alpha_i + \beta_i(R_{m,t} - R_{b,t}) + \epsilon_{i,t}$$

- α_i is alpha.
 - CAPM predicts that $\alpha_i = 0$ for all assets.
 - If $\alpha_i > 0$, average return is too high relative to risk.
- The residual $\epsilon_{i,t}$ represents idiosyncratic risk that is uncorrelated with the market.
- Typically tested on portfolios rather than individual stocks.

A statistics refresher

- Linear regression equation:

$$y_t = \alpha + \beta x_t + \epsilon_t$$

where you observe time-series data for $t = 1, \dots, T$.

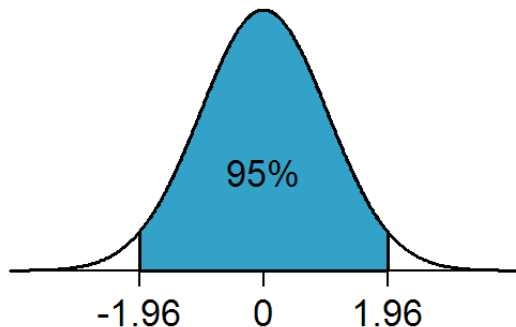
- Define sample means as

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t, \quad \bar{x} = \frac{1}{T} \sum_{t=1}^T x_t$$

- Ordinary least squares estimator is

$$\hat{\beta} = \frac{\text{Cov}(x_t, y_t)}{\text{Var}(x_t)} = \frac{\sum_{t=1}^T (x_t - \bar{x})(y_t - \bar{y})}{\sum_{t=1}^T (x_t - \bar{x})^2}$$
$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

- Standard errors $\sigma(\hat{\beta})$ and $\sigma(\hat{\alpha})$ measure uncertainty around point estimates.
- t -statistic is the ratio of point estimate to standard error: $\hat{\beta}/\sigma(\hat{\beta})$ and $\hat{\alpha}/\sigma(\hat{\alpha})$.
- In large samples, t -statistics are normally distributed.
- To reject the null hypothesis $\alpha = 0$ at the 95% significance level, the t -statistic must be greater than 1.96 in absolute value.



Linear regression in Excel

- Use the regression function in the Data Analysis ToolPak.
 - First install the Data Analysis ToolPak add-in.
 - See <https://www.excel-easy.com/examples/regression.html>

CAPM for individual stocks

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Ticker	Company	Beta	Alpha	R-Square	Residual Std Dev	Standard Error Beta	Standard Error Alpha	Adjusted Beta
MSFT	Microsoft	0.856	0.008	0.224	0.056	0.209	0.007	0.904
KO	Coca-Cola	0.406	0.004	0.129	0.037	0.138	0.005	0.604
XOM	ExxonMobil	0.876	-0.003	0.460	0.033	0.125	0.004	0.917
GE	General Electric	1.093	0.004	0.467	0.041	0.153	0.005	1.062
BA	Boeing	0.944	0.009	0.350	0.045	0.169	0.006	0.963
NEM	Newmont Mining	0.316	-0.016	0.011	0.107	0.400	0.014	0.544
PFE	Pfizer	0.837	0.007	0.432	0.034	0.126	0.004	0.891
F	Ford	1.326	-0.010	0.394	0.058	0.216	0.008	1.217
DIS	Walt Disney	1.352	0.009	0.617	0.037	0.140	0.005	1.235
BP	BP	1.533	-0.012	0.584	0.045	0.170	0.006	1.355
MCD	McDonald's	0.455	0.006	0.186	0.033	0.125	0.004	0.637
WMT	Walmart	0.209	0.003	0.023	0.048	0.179	0.006	0.473
INTC	Intel	0.831	0.006	0.221	0.055	0.205	0.007	0.887
BAC	Bank of America	1.681	-0.005	0.361	0.078	0.294	0.010	1.454
VZ	Verizon	0.237	0.007	0.037	0.042	0.158	0.006	0.492
X	U.S. Steel	1.844	-0.039	0.255	0.111	0.414	0.015	1.563
AMZN	Amazon	0.866	0.018	0.142	0.075	0.279	0.010	0.911
GOOG	Alphabet (Google)	0.799	0.011	0.166	0.063	0.235	0.008	0.866
AVERAGE		0.915	0.000	0.281	0.055	0.207	0.007	0.943
STD DEVIATION		0.488	0.013	0.186	0.024	0.088	0.003	0.325

Table 8.2

Index model estimates: Regressions of excess stock returns on the excess return of the broad market index over 60 months, 2011–2015

Source: Bodie, Kane, and Marcus (2018, Table 8.2)

Market betas for industry portfolios

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Industry	Beta	Adjustment Factor
Agriculture	0.99	−0.140
Drugs and medicine	1.14	−0.099
Telephone	0.75	−0.288
Energy utilities	0.60	−0.237
Gold	0.36	−0.827
Construction	1.27	0.062
Air transport	1.80	0.348
Trucking	1.31	0.098
Consumer durables	1.44	0.132

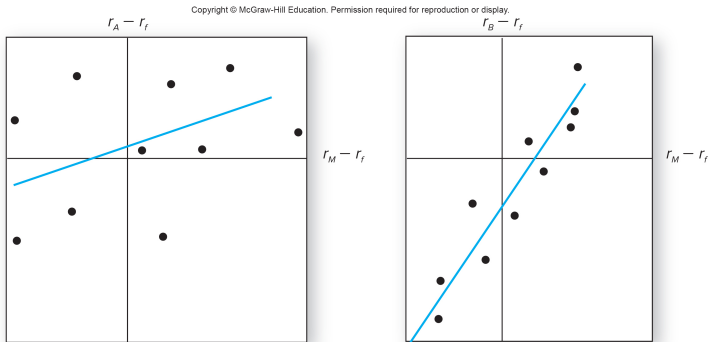
Table 8.3

Industry betas and
adjustment factors

Source: Bodie, Kane, and Marcus (2018, Table 8.3)

Low vs. high market beta

- CAPM predicts that stocks with higher market beta have higher expected returns.
- Test on portfolios sorted by market beta.

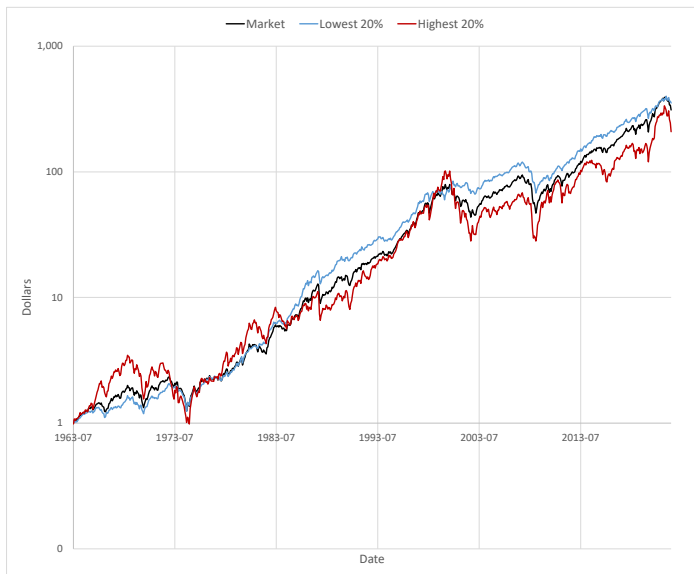


Source: Bodie, Kane, and Marcus (2018, Figure 8.1)

Constructing portfolios sorted by market beta

1. Using monthly returns for the previous 60 months, estimate market beta for each stock.
2. Sort stocks into 5 portfolios (i.e., cutoff at each 20th percentile).
3. Compute returns on these portfolios over the subsequent year.
4. Rebalance annually.

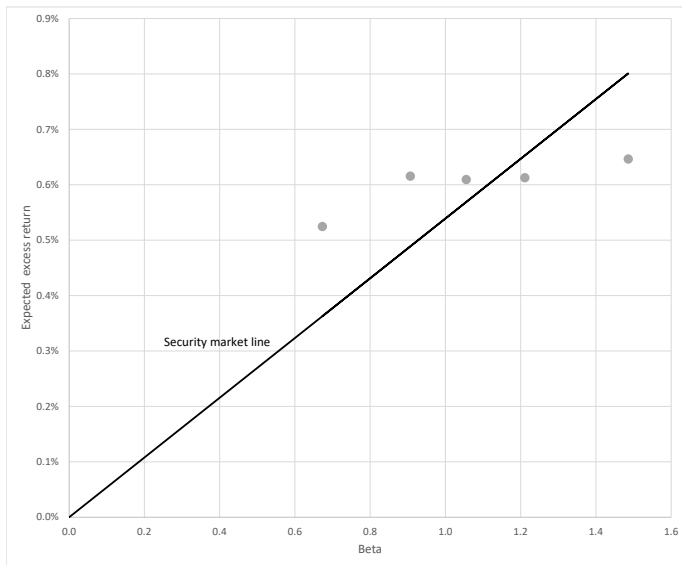
Cumulative returns on low vs. high market beta



Example 1: Portfolios sorted by market beta

Statistic	Market	Portfolio				
		1	2	3	4	5
Mean (%)	0.54	0.52	0.62	0.61	0.61	0.65
SD (%)	4.37	3.47	4.15	4.76	5.54	7.08
Sharpe ratio	0.12	0.15	0.15	0.13	0.11	0.09
Beta		0.67	0.91	1.06	1.21	1.49
Alpha (%)		0.16	0.12	0.04	0.00	-0.16
Alpha (t -stat)		2.33	2.47	0.82	-0.01	-1.54

Alpha on portfolios sorted by market beta



Short riskless asset to increase beta

- Portfolio 1:

$$\mathbb{E}[R_1 - R_b] = 0.16\% + 0.67 \times \mathbb{E}[R_m - R_b]$$

- Invest $1/0.67 = 1.49$ in portfolio 1 and borrow -0.49 at the riskless rate.
 - Beta is

$$1.49 \times 0.67 = 1$$

- Expected return is

$$R_b + 1.49 \times \mathbb{E}[R_1 - R_b] = 0.24\% + \mathbb{E}[R_m]$$

- Same risk as the market but higher expected return.
 - Arbitrage if the CAPM is true.

Long riskless asset to decrease beta

- Portfolio 5:

$$\mathbb{E}[R_5 - R_b] = -0.16\% + 1.49 \times \mathbb{E}[R_m - R_b]$$

- Invest $1/1.49 = 0.67$ in portfolio 5 and 0.33 in the riskless asset.

- Beta is

$$0.67 \times 1.49 = 1$$

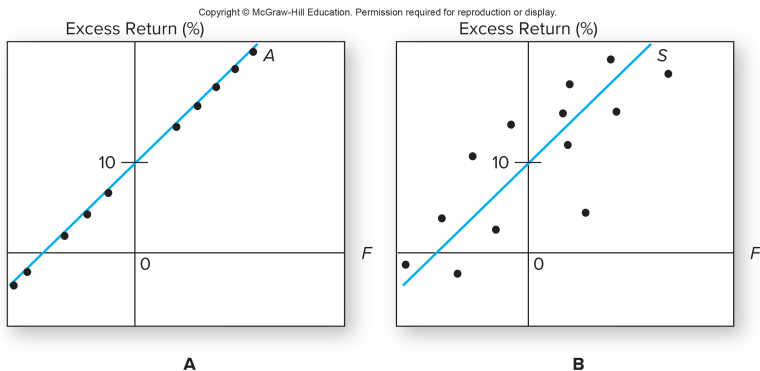
- Expected return is

$$R_b + 0.67 \times \mathbb{E}[R_5 - R_b] = -0.11\% + \mathbb{E}[R_m]$$

- Same risk as the market but lower expected return.

Low vs. high residual variance

- CAPM predicts that residual variance is unrelated to expected returns.
- Test on portfolios sorted by residual variance.



Source: Bodie, Kane, and Marcus (2018, Figure 10.1)

Constructing portfolios sorted by residual variance

1. Using daily returns for the previous 60 days, estimate residual variance for each stock.
2. Sort stocks into 5 portfolios (i.e., cutoff at each 20th percentile).
3. Compute returns on these portfolios over the subsequent month.
4. Rebalance monthly.

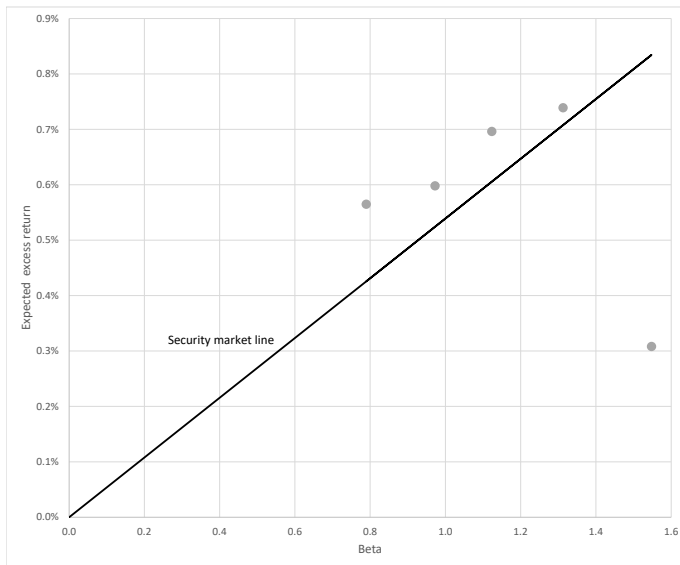
Cumulative returns on low vs. high residual variance



Example 2: Portfolios sorted by residual variance

Statistic	Market	Portfolio				
		1	2	3	4	5
Mean (%)	0.54	0.56	0.60	0.70	0.74	0.31
SD (%)	4.37	3.66	4.43	5.13	6.05	7.72
Sharpe ratio	0.12	0.15	0.14	0.14	0.12	0.04
Beta		0.79	0.97	1.12	1.31	1.55
Alpha (%)		0.16	0.07	0.07	0.04	-0.56
Alpha (<i>t</i> -stat)		3.16	1.44	1.35	0.53	-3.85

Alpha on portfolios sorted by residual variance



Summary

- Low beta stocks have abnormally high returns relative to risk.
- Stock's residual variance predicts returns.
 - Low $\Rightarrow$ Abnormally high returns.
 - High $\Rightarrow$ Abnormally low returns.
- A strategy that sorts stocks on variance (instead of residual variance) works similarly.
- Low volatility strategies are widely available in the mutual fund industry.
- Instead of investing in the market, invest in low volatility stocks for higher returns and lower risk!

iShares MSCI USA Min Vol Factor ETF (USMV)

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Page 10 of 16

Holdings



Portfolio Holdings	No.	%
Equity Holding	173	96.11
Bond Holding	0	0.00
Other Holding	7	3.89

% Assets in Top 10 Holdings

Reported Turnover %
23.00
As of 07/31/21

Women Directors %

32

Women Executives %

26

Holdings	% Portfolio Weight	First Bought	Market Value USD as of Sep 29, 2022	1-Year Return	Forward P/E	Equity Star Rating	Economic Moat	ESG Risk Rating Assessment	Sector
Vertex Pharmaceuticals Inc	1.81	May 29, 2020	496,309,469	59.62	18.69	★★★	Narrow	@@@	Healthcare
T-Mobile US Inc	1.69	May 31, 2017	463,131,845	5.02	18.18	★★★★	Narrow	@@@	Communication Services
Republic Services Inc	1.68	Dec 03, 2012	460,087,997	14.87	25.77	★	Wide	@@@	Industrials
Waste Management Inc	1.67	Oct 19, 2011	458,905,075	8.95	24.88	★★	Wide	@@@	Industrials
Waste Connections Inc	1.61	Nov 30, 2016	441,157,534	8.04	22.94	★	Wide	@@@	Industrials
Cisco Systems Inc	1.54	Jun 03, 2013	421,063,940	-23.76	11.33	★★★★	Wide	@@@	Technology
Johnson & Johnson	1.49	Oct 19, 2011	409,574,565	3.86	15.43	★★★	Wide	@@@	Healthcare
PepsiCo Inc	1.49	Oct 19, 2011	409,550,705	11.50	22.37	★★★	Wide	@@@	Consumer Defensive
Merck & Co Inc	1.49	Oct 19, 2011	407,374,002	18.33	11.48	★★★	Wide	@@@	Healthcare
Texas Instruments Inc	1.47	Nov 30, 2017	401,999,692	-17.08	17.33	★★★	Wide	@@@	Technology

Page 1 of 10 < > Viewing 1 to 10 of 100

Holdings as of Sep 29, 2022 | The top 100 largest holdings are available for display across Equity, Bond and Other.

Source: Morningstar Direct

What explains the failure of CAPM?

- SEC disclaimer: “Past performance does not necessarily predict future results”.
- Maybe the CAPM will work better in the future.
- Reasons for the failure of CAPM.
 - If investors with low risk aversion cannot borrow, they bid up the price of high beta stocks.
 - Fund managers try to beat their benchmarks by tilting their portfolios toward high beta stocks.
- CAPM builds on the portfolio separation theorem.
 - All investors should have portfolios on the capital allocation line. But recall evidence from Sweden.

Roll's critique

- The market portfolio in the CAPM is not just the US stock market but includes bonds, international stocks, real estate, private equity, and so on.
- Impossible to test the CAPM because the market return is hard to measure.
- Maybe the CAPM does work, but we do not have a good empirical proxy for the market portfolio.

Next steps

- There are other types of anomalies that the CAPM does not explain.
- The most famous of which are size and value investing.
- Multi-factor models that generalize the CAPM and allow more sources of systematic risk to be priced.